HEATER, VENTILATION AND AIR CONDITIONING

id070000126400

Abbreviation

AAS	Active Adaptive Shift
ABS	Antilock Brake System
ABDC	After Bottom Dead Center
ACC	Accessories
AFS	Adaptive Front lighting System
ALC	Auto Level Control
ALR	Automatic Locking Retractor
ATDC	After Top Dead Center
ATF	Automatic Transaxle Fluid
ATX	Automatic Transaxle
BBDC	Before Bottom Dead Center
BCM	Body Control Module
BDC	Bottom Dead Center
BTDC	Before Top Dead Center
CAN	Controller Area Network
CCM	Comprehensive Component Monitor
CKP	Crankshaft Position
CM	Control Module
CMDTC	Continuous Memory Diagnostic Trouble Code
CMP	Camshaft Position
CMU	Connectivity Master Unit
CPU	Central Processing Unit
DAB	Digital Audio Broadcasting
DC	Drive Cycle
DEF	Defroster
DRSS	Distance Recognition Support System
DSC	Dynamic Stability Control
EBD	Electronic Brakeforce Distribution
EEPROM	Electrically Erasable Programmable Read-Only Memory
EHPAS	Electro Hydraulic Power Assist Steering
ELR	Emergency Locking Retractor
EPS	Electric Power Steering
ESS	Emergency Stop signal System
ESU	Electrical Supply Unit
EX	Exhaust
FBCM	Front Body Control Module
FSC	Forward Sensing Camera
GPS	Global Positioning System
HBC	High Beam Control
HF/TEL	Hands-Free Telephone
HI	High
HLA	Hydraulic Lash Adjuster
HP	High Pressure
HS	High Speed
HU	Hydraulic Unit
IDEVA	Intake stroke EGR using Double Exhaust Valve Actuation system
IDS	Integrated Diagnostic Software
IG	Ignition
IN	Intake
INT	Intermittent
KOEO	Key On Engine Off
KOER	Key On Engine Running
LCD	Liquid Crystal Display
LDWS	Lane Departure Warning System
LED	Light Emitting Diode
LF	Left Front
LH	Left Hand
L.H.D.	Left Hand Drive

LO	Low
LP	Low Pressure
LR	Left Rear
M	Motor
MAX	Maximum
MIN	Minimum
MRCC	Mazda Radar Cruise Control
MS	Middle speed
MTX	Manual Transaxle
NVH	Noise, Vibration, Harshness
OCV	Oil Control Valve
ODDTC	On-demand Diagnostic Trouble Code
PAD	Passenger Air Bag Deactivation
PCV	Positive Crankcase Ventilation
PDS	Portable Diagnostic Software
PID	Parameter Identification
POWER MOS FET	Power Metal Oxide Semiconductor Field Effect Transistor
PSD	Power Sliding Door
P/W CM	Power Window Control Module
PTC	Positive Temperature Coefficient
RBCM	Rear Body Control Module
RCTA	Rear Crossing Traffic Alert
RDP	Run Dry Prevention
RDS	Radio Data System
REC	Recirculate
RES	Rear Entertainment System
RF	Right Front
RH	Right Hand
R.H.D.	Right Hand Drive
RR	Right Rear
SAS	Sophisticated Air Bag Sensor
SST	Special Service Tool
SW	Switch
TAU	Tuner and Amp Unit
TCS	Traction Control System
TDC	Top Dead Center
TFT	Transaxle Fluid Temperature
TNS	Tail Number Side Lights
TPMS	Tire Pressure Monitoring System
USB	Universal Serial Bus
VBC	Variable Boost Control
VENT	Ventilation
WGN	Wagon
W/M	Workshop Manual
1GR	First Gear
2GR	Second Gear
2WD	2-Wheel Drive
3GR	Third Gear
4GR	Fourth Gear
4SD	4 Door Sedan
4WD	4-Wheel Drive
5GR	Fifth Gear
5HB	5 Door Hatchback
6GR	Sixth Gear

Features

- An A/C unit and separate blower unit have been adopted to the blower system.
- An A/C pressure sensor has been adopted in which refrigerant pressure is converted to a linear electrical signal and precise information is transmitted.
- A sub-cooling type condenser with an integrated condenser and receiver/drier have been adopted.(SKYACTIV-G 1.3, SKYACTIV-G 1.5)

- To improve the heating performance temporarily under cold diesel engine conditions, a PTC heater which uses electricity as the heat source as been adopted. (SKYACTIV-D 1.5 (With PTC heater))

Specifications
Basic system (SKYACTIV-G 1.3, SKYACTIV-G 1.5)

Item				Specification
Heating capacity			(kW {kcal/h})	5.2 {4,471}
Cooling capacity			(kW {kcal/h})	4.7 {4,041}
Temperature control				Reheat full air mix type
Blower motor	Fan type			Sirocco fan
	During heater operation	Airflow volume	(m^3/h)	240
		Electricity consumption	(W)	150
	During air conditioner operation	Airflow volume	(m^3/h)	400
		Electricity consumption	(W)	200
Refrigerant	Type			HFC-134a (R-134a) HFO-1234yf (R-1234yf)
	Regular amount (approx. quantity)		(g {oz})	395—445 {14.0—15.6} (HFC-134a (R-134a)) 335—385 {11.9—13.5} (HFO-1234yf (R-1234yf))
A/C compressor	Type			Vane-rotary
	Discharge capacity		(ml {cc, fl oz})	90 {90, 3.0}
	Max. allowable speed		(rpm)	6,100
	Lube oil	Type		ZXL200PG (HFC-134a (R-134a)) VC200YF (HFO-1234yf (R-1234yf))
		Sealed volume (approx. quantity)	(ml {cc, fl oz})	110 {110, 3.72} (HFC-134a (R-134a)) 110 {110, 3.72} (HFO-1234yf (R-1234yf))
Magnetic clutch	Magnetic clutch clearance		(mm {in})	0.3—0.6 {0.012—0.023}
	Electricity consumption (during air conditioner operation)		(W)	45
Condenser	Type			Multi-flow (sub-cooling type)
	Radiated heat		(kW {kcal/h})	6.5 {5,589}
	Receiver/drier capacity		(ml {cc, fl oz})	160 {160, 5.41}
	Desiccant			Synthetic zeolite
Expansion valve	Type			External equalizing type
Evaporator	Type			Multi-flow type

Note
- The refrigerant used for the refrigerant system differs depending on the country. When draining or adding the refrigerant, verify the appropriate refrigerant type and specified amount of refrigerant from the caution label.

Basic system (SKYACTIV-D 1.5)

Item				Specification
Heating capacity			(kW {kcal/h})	5.2 {4,471}
Cooling capacity			(kW {kcal/h})	4.7 {4,041}
Temperature control				Reheat full air mix type
Blower motor	Fan type			Sirocco fan
	During heater operation	Airflow volume	(m^3/h)	240
		Electricity consumption	(W)	150
	During air conditioner operation	Airflow volume	(m^3/h)	400
		Electricity consumption	(W)	200

Item				Specification
Refrigerant	Type			HFC-134a (R-134a) HFO-1234yf (R-1234yf)
	Regular amount (approx. quantity)		(g {oz})	395—445 {14.0—15.6} (HFC-134a (R-134a)) 365—415 {12.9—14.6} (HFO-1234yf (R-1234yf))
A/C compressor	Type			Vane-rotary
	Discharge capacity		(ml {cc, fl oz})	83 {83, 2.8}
	Max. allowable speed		(rpm)	6,100
	Lube oil	Type		DH-PR (HFC-134a (R-134a)) YR20 (HFO-1234yf (R-1234yf))
		Sealed volume (approx. quantity)	(ml {cc, fl oz})	140 {140, 4.73} (HFC-134a (R-134a)) 130 {130, 4.40} (HFO-1234yf (R-1234yf))
Magnetic clutch	Magnetic clutch clearance		(mm {in})	0.3—0.6 {0.012—0.023}
	Electricity consumption (during air conditioner operation)		(W)	40
Condenser	Type			Multi-flow (sub-cooling type)
	Radiated heat		(kW {kcal/h})	6.1 {5,245}
	Receiver/drier capacity		(ml {cc, fl oz})	180 {180, 6.09}
	Desiccant			Synthetic zeolite
Expansion valve	Type			External equalizing type
Evaporator	Type			Multi-flow type
PTC heater	Heating performance		(kW {kcal/h})	1.0 {860}
	Energy consumption when heating(During PTC heater operation)		(W)	1,000

Note

- The refrigerant used for the refrigerant system differs depending on the country. When draining or adding the refrigerant, verify the appropriate refrigerant type and specified amount of refrigerant from the caution label.

Control system (full-auto air conditioner)

Item		Specification
Sensor	Refrigerant pressure sensor	Capacitance type
	Solar radiation sensor *	Photodiode
	Ambient temperature sensor	Thermistor
	Cabin temperature sensor	
	Evaporator temperature sensor	
Actuator	Air intake actuator	Mechanical lock type
	Air mix actuator	Potentiometer type
	Airflow mode actuator	

* : With solar radiation sensor

Control system (manual air conditioner)

Item		Specification
Sensor	Refrigerant pressure sensor	Capacitance type
	Ambient temperature sensor	Thermistor
	Evaporator temperature sensor	
Actuator	Air intake actuator	Mechanical lock type